DATA 505: Homework Quiz 2

Name: ________________________________

2025-09-16

In R, suppose we have a object `penguins` with the following structure.

```
'data.frame':   344 obs. of  8 variables:
 $ species          : chr  "Adelie" "Adelie" "Adelie" "Adelie" ...
 $ island           : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
 $ bill_length_mm   : num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm    : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int  181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g      : int  3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex              : chr  "male" "female" "female" NA ...
 $ year             : int  2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

Question 1 (3 points)

Which of the following R operations would result in coercing the `species` variable to a `factor` object, and storing the result in the `species` column of the data frame? *(Circle the correct answer.)*

A. `penguins.species <- penguins.species.as_factor()`

B. `as_factor(penguins[["species"]])`

C. `penguins[["species"]] <- as.factor(penguins$species)`

D. There is no need for converting: `species` is already a factor object.

Question 2 (3 points)

Which of the following R operations would result in subsetting the data frame, keeping only those rows with non-missing body mass (`body_mass_g`), and only the `species` and `island` columns? *(Circle the correct answer.)*

A. `penguins[!is.na(penguins$body_mass_g), c("species", "island")]`

B. `penguins.[!is.na(body_mass_g); species, island]`

C. `penguins[[!is.na("body_mass_g)]][c("species", "island")]`

D. `penguins.drop_na("body_mass_g").select(["species", "island"])`

Question 3 (4 points)

```r
summary(penguins$body_mass_g)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
   2700    3550    4050    4202    4750    6300       2 
```

Based on the output above, what is the interquartile range (IQR) of the body mass (`body_mass_g`)? *(Circle the correct answer.)*

A. 4050 (the median)

B. 4202 (the mean)

C. 3600 ($6300 - 2700$)

D. 1200 ($4750 - 3550$)